Assignment 3

① a: Thermal voltage and Diode current

Temperature in kelvin

$T_K = 25 + 273 = 298\,K.$

V_T: $$V_T = \frac{(1.38 \times 10^{-23})(298)}{1.602 \times 10^{-19}}$$

$$V_T = 0.02567\,v\,(25.67\ mV)$$

ⓑ Find the diode current we can use the given values

$I_S = 40\,nA$. $n = 2$, $V_D = 0.5\,V.$

Equation: $I_D = I_S\left(e^{\frac{V_D}{n V_T}} - 1\right)$

Calculate: $I_D = (40 \times 10^{-9})\left(e^{\frac{0.5}{2 \times 0.02567}} - 1\right)$

$$I_D = (40 \times 10^{-19})\left(e^{9.739} - 1\right)$$

$I_D = 0.0006778 A$ or $(0.678 mA)$

② Approx characteristic in the model a forward biased silicon diode has a voltage drop of 0.7 V

- V_D: The voltage across the diode is simply 0.7V.

V_R: The remaining voltage from the 30V source across the 1.5 kΩ resistor.

$$V_R = 30 - 0.7 = 29.3 V.$$

∴ I_D using ohm's law $(I = \frac{V}{R})$

$$I_D = \frac{29.3}{1500} = 0.01953 A.$$

① The diode acts as a perfect swich

with a 0V drop

V_D: The voltage across the diode is 0V

V_R: The entire 30V drops across the resistor. $V_R = 30V$

$$I_D = \frac{30}{1500} = 0.02A \ (20mA)$$

c. yes. the result suggest the ~~do~~ Ideal model povides a very good approximation in this sceanario. Because the applied source voltage (30V) is signisically larger than the diode's built in drop (0.7V), the difference in calculated current is minimal.

Q3 a. The diode is connected to the negative terminal and reversed-biased, it acts as an open circuit.

$$I = 0A$$

(b) The 20V battery has its positive terminal connected to the ground, making the top node -20V. The diode's anode are connected to the ground (0V) through the resistors. Since current flows from higher to lower potential (0V to 20V), both diodes are forward-biased.

Left Branch Current (I_1)

$$0 - 0.7 - I_1(10) = -20 \Rightarrow 10I_1 = 19$$

$$\Rightarrow I_1 = 1.93A.$$

Middle Branch current (I_2)

$$0 - 0.7 - I_2(20) = -20 + 20I_2 = 19.3$$

$$\Rightarrow I_2 = 0.905A$$

Total current (I)

$$I = I_1 + I_2 = 1.93 + 0.905 = 2.895A$$

The current I flows entirely through the outer loop with Ra = 10V battery and I = 10/10 = 1 A

Q.4 circuit a.

The silicon diode D_1 (vdrop) and gated diode L_1, L_2 (vdrop) are in parallel. The silicon diode will have an offset changing the voltage across the parallel block to 0.2v. The gated diode will remain off.

$$V_0 - V_0 = 1 - 0.7 = 0.3V$$

$$T = T_1 = \frac{V_0}{R} = \frac{0.3}{1000} = 0.0003 A \text{ or } 0.3$$

Circuit B.

The ~~eff~~ current flows through the top ~~Si~~ Si diode and then split through 2 parallel Si diodes. The total voltage drop across the diode is

$$V_o = V_o = (16 - 1.9) = 14.6\ V.$$

$$V_{offset} = 14.6 - (-4) = 18.6\ V.$$

$$I = \frac{18.6}{470}$$

$$= 0.03957\ A \text{ as } 3.95\ mA$$